JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Groups (2015–2021)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
50	82 M	150 Min	MCQ • MSQ • NAT (2015–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	21	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	19	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q51 • ID: JAM_2021_Q51) NAT +2 M Neg: Nil (0)

Q. 51 The number of elements of order two in the group S_4 is equal to _____.

Question 2 (JAM 2021 • Q47 • ID: JAM_2021_Q47) NAT +1 M Neg: Nil (0)

JAM 2021 MATHEMATICS - MA

Q. 47 The number of group homomorphisms from the group $\mathbb{Z}_4$ to the group S_3 is _____.

Question 3 (JAM 2021 • Q41 • ID: JAM_2021_Q41) NAT +1 M Neg: Nil (0)

Q. 41 The number of cycles of length 4 in S_6 is _____.

Question 4 (JAM 2021 • Q36 • ID: JAM_2021_Q36) MSQ +2 M Neg: Nil (0)

Q. 36 Let G be a finite group of order 28. Assume that G contains a subgroup of order 7. Which of the following statements is/are true?

(A) G contains a unique subgroup of order 7.

(B) G contains a normal subgroup of order 7.

(C) G contains no normal subgroup of order 7.

(D) G contains at least two subgroups of order 7.

Question 5 (JAM 2021 • Q28 • ID: JAM_2021_Q28) MCQ +2 M Neg: −0.67

Q. 28 Let G be a finite abelian group of odd order. Consider the following two statements:

I. The map $f : G \to G$ defined by $f(g) = g^2$ is a group isomorphism.

II. The product $\prod_{g \in G} g = e$.

(A) Both I and II are TRUE.

(B) I is TRUE but II is FALSE.

(C) II is TRUE but I is FALSE.

(D) Neither I nor II is TRUE.

MA 9 / 17

Question 6 (JAM 2021 • Q27 • ID: JAM_2021_Q27) MCQ +2 M Neg: −0.67

Q. 27 Let g be an element of S_7 such that g commutes with the element $(2, 6, 4, 3)$. The number of such g is

(A) 6.

(B) 4.

(C) 24.

(D) 48.

Question 7 (JAM 2021 • Q24 • ID: JAM_2021_Q24) MCQ +2 M Neg: −0.67

Q. 24 Which one of the following statements is true?

(A) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{R}, +)$.

(B) $(\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

(C) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}/2\mathbb{Z}, +)$.

(D) $(\mathbb{Q}/\mathbb{Z}, +)$ is isomorphic to $(\mathbb{Q}, +)$.

MA 8 / 17

Question 8 (JAM 2021 • Q20 • ID: JAM_2021_Q20) **MCQ** **+2 M** **Neg: −0.67**

Q. 20 Which one of the following statements is true?

(A) Exactly half of the elements in any even order subgroup of S_5 must be even permutations.

(B) Any abelian subgroup of S_5 is trivial.

(C) There exists a cyclic subgroup of S_5 of order 6.

(D) There exists a normal subgroup of S_5 of index 7.

Question 9 (JAM 2021 • Q14 • ID: JAM_2021_Q14) **MCQ** **+2 M** **Neg: −0.67**

Q. 14 Consider the following statements.

I. The group $(\mathbb{Q}, +)$ has no proper subgroup of finite index.

II. The group $(\mathbb{C} \setminus \{0\}, \cdot)$ has no proper subgroup of finite index.

Which one of the following statements is true?

(A) Both I and II are TRUE.

(B) I is TRUE but II is FALSE.

(C) II is TRUE but I is FALSE.

(D) Neither I nor II is TRUE.

MA 5 / 17

Question 10 (JAM 2021 • Q6 • ID: JAM_2021_Q6) **MCQ** +1 M **Neg: −0.33**

Q. 6 How many elements of the group $\mathbb{Z}_{50}$ have order 10?

(A) 10 (B) 4 (C) 5 (D) 8

Question 11 (JAM 2020 • Q60 • ID: JAM_2020_Q60) NAT +2 M Neg: Nil (0)

Q. 60 Suppose that G is a group of order 57 which is NOT cyclic. If G contains a unique subgroup H of order 19, then for any $g \notin H$, $o(g)$ is ________.

END OF THE QUESTION PAPER

MA 17 / 17

Question 12 (JAM 2020 • Q50 • ID: JAM_2020_Q50) NAT +1 M Neg: Nil (0)

Q. 50 Let $\phi : S_3 \to S^1$ be a non-trivial non-injective group homomorphism. Then, the number of elements in the kernel of ϕ is ________.

MA 15 / 17

Question 13 (JAM 2020 • Q40 • ID: JAM_2020_Q40) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 40 Let G be a group with identity e. Let H be an abelian non-trivial proper subgroup of G with the property that $H \cap gHg^{-1} = \{e\}$ for all $g \notin H$.

If $K = \{g \in G : gh = hg \text{ for all } h \in H\}$, then

(A) K is a proper subgroup of H

(B) H is a proper subgroup of K

(C) $K = H$

(D) there exists no abelian subgroup $L \subseteq G$ such that K is a proper subgroup of L

MA 13 / 17

Question 14 (JAM 2020 • Q30 • ID: JAM_2020_Q30) MCQ +2 M Neg: −0.67

Q. 30 Let $F = \{\omega \in \mathbb{C} : \omega^{2020} = 1\}$. Consider the groups

$$G = \left\{ \begin{pmatrix} \omega & z \\ 0 & 1 \end{pmatrix} : \omega \in F, z \in \mathbb{C} \right\}$$

and

$$H = \left\{ \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} : z \in \mathbb{C} \right\}$$

under matrix multiplication. Then the number of cosets of H in G is

(A) 1010 (B) 2019 (C) 2020 (D) infinite

Question 15 (JAM 2020 • Q29 • ID: JAM_2020_Q29) MCQ +2 M Neg: −0.67

Q. 29 Let $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ be the circle group under multiplication and $i = \sqrt{-1}$. Then the set $\{\theta \in \mathbb{R} : \langle e^{i2\pi\theta} \rangle$ is infinite$\}$ is

(A) empty

(B) non-empty and finite

(C) countably infinite

(D) uncountable

Question 16 (JAM 2020 • Q10 • ID: JAM_2020_Q10) MCQ +1 M Neg: −0.33

Q. 10 Consider the following group under matrix multiplication:

$$H = \left\{ \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} : p, q, r \in \mathbb{R} \right\}.$$

Then the center of the group is isomorphic to

(A) $(\mathbb{R} \setminus \{0\}, \times)$

(B) $(\mathbb{R}, +)$

(C) $(\mathbb{R}^2, +)$

(D) $(\mathbb{R}, +) \times (\mathbb{R} \setminus \{0\}, \times)$

Q. 11 – Q. 30 carry two marks each.

Question 17 (JAM 2019 • Q55 • ID: JAM_2019_Q55) NAT +2 M Neg: Nil (0)

Q.55 The number of elements in the set $\{x \in S_3 : x^4 = e\}$, where e is the identity element of the permutation group S_3, is ___________

Question 18 (JAM 2019 • Q53 • ID: JAM_2019_Q53) NAT +2 M Neg: Nil (0)

Q.53 Let $G = \{n \in \mathbb{N} : n \leq 55, \ \gcd(n, 55) = 1\}$ be the group under multiplication modulo 55. Let $x \in G$ be such that $x^2 = 26$ and $x > 30$. Then x is equal to____________

Question 19 (JAM 2019 • Q41 • ID: JAM_2019_Q41) NAT +1 M Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 Let x be the 100-cycle $(1 \ \ 2 \ \ 3 \ \cdots \ 100)$ and let y be the transposition $(49 \ \ 50)$ in the permutation group S_{100}. Then the order of xy is ______

Question 20 (JAM 2019 • Q32 • ID: JAM_2019_Q32) MSQ +2 M Neg: Nil (0)

Q.32 Let G be a nonabelian group, $y \in G$, and let the maps f, g, h from G to itself be defined by

$$f(x) = yxy^{-1}, \ g(x) = x^{-1} \ \text{and} \ h = g \circ g.$$

Then

(A) g and h are homomorphisms and f is not a homomorphism
(B) h is a homomorphism and g is not a homomorphism
(C) f is a homomorphism and g is not a homomorphism
(D) f, g and h are homomorphisms

Question 21 (JAM 2019 • Q31 • ID: JAM_2019_Q31) MSQ +2 M Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.

Q.31 Let G be a noncyclic group of order 4. Consider the statements I and II:

I. There is NO injective (one-one) homomorphism from G to $\mathbb{Z}_8$
II. There is NO surjective (onto) homomorphism from $\mathbb{Z}_8$ to G

Then

(A) I is true
(B) I is false
(C) II is true
(D) II is false

Question 22 (JAM 2019 • Q11 • ID: JAM_2019_Q11) MCQ +2 M Neg: −0.67

Q.11 Let H and K be subgroups of $\mathbb{Z}_{144}$. If the order of H is 24 and the order of K is 36, then the order of the subgroup $H \cap K$ is

(A) 3 (B) 4 (C) 6 (D) 12

Question 23 (JAM 2018 • Q48 • ID: JAM_2018_Q48) NAT +1 M Neg: Nil (0)

Q.48 Let A_6 be the group of even permutations of 6 distinct symbols. Then the number of elements of order 6 in A_6 is ________________

Question 24 (JAM 2018 • Q41 • ID: JAM_2018_Q41) NAT +1 M Neg: Nil (0)

Q. 41 – Q. 50 carry one mark each.

Q.41 The order of the element $(1\ 2\ 3)\,(2\ 4\ 5)\,(4\ 5\ 6)$ in the group S_6 is ________________

Question 25 (JAM 2018 • Q35 • ID: JAM_2018_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ denote the group of non-zero complex numbers under multiplication. Suppose $Y_n = \{z \in \mathbb{C} \mid z^n = 1\}$, $n \in \mathbb{N}$. Which of the following is (are) subgroup(s) of $\mathbb{C}^*$?

(A) $\bigcup_{n=1}^{100} Y_n$ (B) $\bigcup_{n=1}^{\infty} Y_{2^n}$ (C) $\bigcup_{n=100}^{\infty} Y_n$ (D) $\bigcup_{n=1}^{\infty} Y_n$

Question 26 (JAM 2018 • Q33 • ID: JAM_2018_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Suppose f, g, h are permutations of the set $\{\alpha, \beta, \gamma, \delta\}$, where

f interchanges α and β but fixes γ and δ,

g interchanges β and γ but fixes α and δ,

h interchanges γ and δ but fixes α and β.

Which of the following permutations interchange(s) α and δ but fix(es) β and γ?

(A) $f \circ g \circ h \circ g \circ f$ (B) $g \circ h \circ f \circ h \circ g$ (C) $g \circ f \circ h \circ f \circ g$ (D) $h \circ g \circ f \circ g \circ h$

Question 27 (JAM 2018 • Q28 • ID: JAM_2018_Q28) MCQ +2 M Neg: −0.67

Q.28 Consider the group $\mathbb{Z}^2 = \{(a, b) | \, a, b \in \mathbb{Z}\}$ under component-wise addition. Then which of the following is a subgroup of $\mathbb{Z}^2$?

(A) $\{(a, b) \in \mathbb{Z}^2 | \, ab = 0\}$

(B) $\{(a, b) \in \mathbb{Z}^2 | \, 3a + 2b = 15\}$

(C) $\{(a, b) \in \mathbb{Z}^2 | \, 7 \text{ divides } ab\}$

(D) $\{(a, b) \in \mathbb{Z}^2 | \, 2 \text{ divides } a \text{ and } 3 \text{ divides } b\}$

Question 28 (JAM 2018 • Q26 • ID: JAM_2018_Q26) MCQ +2 M Neg: −0.67

Q.26 Let H be the quotient group $\mathbb{Q}/\mathbb{Z}$. Consider the following statements.

I. Every cyclic subgroup of H is finite.

II. Every finite cyclic group is isomorphic to a subgroup of H.

Which one of the following holds?

(A) I is TRUE but II is FALSE (B) II is TRUE but I is FALSE

(C) both I and II are TRUE (D) neither I nor II is TRUE

Question 29 (JAM 2018 • Q25 • ID: JAM_2018_Q25) MCQ +2 M Neg: −0.67

Q.25 Let G be a group satisfying the property that $f: G \to \mathbb{Z}_{221}$ is a homomorphism implies $f(g) = 0,\ \forall g \in G$. Then a possible group G is

(A) $\mathbb{Z}_{21}$ (B) $\mathbb{Z}_{51}$ (C) $\mathbb{Z}_{91}$ (D) $\mathbb{Z}_{119}$

MA 6/12

Question 30 (JAM 2018 • Q1 • ID: JAM_2018_Q1) MCQ +1 M Neg: −0.33

Q. 1 – Q.10 carry one mark each.

Q.1 Which one of the following is TRUE?

(A) $\mathbb{Z}_n$ is cyclic if and only if n is prime

(B) Every proper subgroup of $\mathbb{Z}_n$ is cyclic

(C) Every proper subgroup of S_4 is cyclic

(D) If every proper subgroup of a group is cyclic, then the group is cyclic

Question 31 (JAM 2017 • Q54 • ID: JAM_2017_Q54) NAT +2 M Neg: Nil (0)

Q.54 The maximum order of a permutation σ in the symmetric group S_{10} is ________

Question 32 (JAM 2017 • Q50 • ID: JAM_2017_Q50) NAT +1 M Neg: Nil (0)

Q.50 The number of subgroups of $\mathbb{Z}_7 \times \mathbb{Z}_7$ of order 7 is ________

Question 33 (JAM 2017 • Q43 • ID: JAM_2017_Q43) NAT +1 M Neg: Nil (0)

Q.43 Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 7 & 8 & 6 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 1 & 7 & 6 & 8 & 2 \end{pmatrix}$ in S_8. The number of $\eta \in S_8$ such that $\eta^{-1}\sigma\eta = \tau$ is equal to ________

Question 34 (JAM 2017 • Q42 • ID: JAM_2017_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let G be a subgroup of $GL_2(\mathbb{R})$ generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$. Then the order of G is

____

Question 35 (JAM 2017 • Q35 • ID: JAM_2017_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let G be a group of order 20 in which the conjugacy classes have sizes $1, 4, 5, 5, 5$. Then which of the following is/are TRUE?

(A) G contains a normal subgroup of order 5

(B) G contains a non-normal subgroup of order 5

(C) G contains a subgroup of order 10

(D) G contains a normal subgroup of order 4

Question 36 (JAM 2017 • Q31 • ID: JAM_2017_Q31) MSQ +2 M Neg: Nil (0)

Q.31 For $\alpha, \beta \in \mathbb{R}$, define the map $\varphi_{\alpha,\beta} : \mathbb{R} \to \mathbb{R}$ by $\varphi_{\alpha,\beta}(x) = \alpha x + \beta$. Let

$$G = \{\varphi_{\alpha,\beta} \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

For $f, g \in G$, define $g \circ f \in G$ by $(g \circ f)(x) = g(f(x))$. Then which of the following statements is/are TRUE?

(A) The binary operation $\circ$ is associative

(B) The binary operation $\circ$ is commutative

(C) For every $(\alpha, \beta) \in \mathbb{R}^2$, $\alpha \neq 0$ there exists $(a, b) \in \mathbb{R}^2$ such that $\varphi_{\alpha,\beta} \circ \varphi_{a,b} = \varphi_{1,0}$

(D) $(G, \circ)$ is a group

Question 37 (JAM 2017 • Q3 • ID: JAM_2017_Q3) MCQ +1 M Neg: −0.33

Q.3 The number of generators of the additive group $\mathbb{Z}_{36}$ is equal to

(A) 6 (B) 12 (C) 18 (D) 36

Question 38 (JAM 2016 • Q58 • ID: JAM_2016_Q58) NAT +2 M Neg: Nil (0)

Q.58 Let H denote the group of all 2×2 invertible matrices over $\mathbb{Z}_5$ under usual matrix multiplication. Then the order of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ in H is ____________

Question 39 (JAM 2016 • Q50 • ID: JAM_2016_Q50) NAT +1 M Neg: Nil (0)

Q.50 Let G be a cyclic group of order 12. Then the number of non-isomorphic subgroups of G is ____________

Q. 51 – Q. 60 carry two marks each.

Question 40 (JAM 2016 • Q38 • ID: JAM_2016_Q38) MSQ +2 M Neg: Nil (0)

Q.38 Let G be a finite group and $o(G)$ denotes its order. Then which of the following statement(s) is(are) TRUE?

(A) G is abelian if $o(G) = pq$ where p and q are distinct primes
(B) G is abelian if every non identity element of G is of order 2
(C) G is abelian if the quotient group $\frac{G}{Z(G)}$ is cyclic, where $Z(G)$ is the center of G
(D) G is abelian if $o(G) = p^3$, where p is prime

Question 41 (JAM 2016 • Q29 • ID: JAM_2016_Q29) MCQ +2 M Neg: −0.67

Q.29 In the permutation group S_n $(n \geq 5)$, if H is the smallest subgroup containing all the 3-cycles, then which one of the following is TRUE?

(A) Order of H is 2
(B) Index of H in S_n is 2
(C) H is abelian
(D) $H = S_n$

MA 7/13

Question 42 (JAM 2016 • Q28 • ID: JAM_2016_Q28) MCQ +2 M Neg: −0.67

Q.28 The number of group homomorphisms from the cyclic group $\mathbb{Z}_4$ to the cyclic group $\mathbb{Z}_7$ is

(A) 7 (B) 3 (C) 2 (D) 1

Question 43 (JAM 2016 • Q4 • ID: JAM_2016_Q4) MCQ +1 M Neg: −0.33

Q.4 Let σ be an element of the permutation group S_5. Then the maximum possible order of σ is

(A) 5 (B) 6 (C) 10 (D) 15

Question 44 (JAM 2015 • Q52 • ID: JAM_2015_Q52) NAT +2 M Neg: Nil (0)

Q.12 Suppose G is a cyclic group and $\sigma, \tau \in G$ are such that order(σ) = 12 and order(τ) = 21. Then the order of the smallest group containing σ and τ is ______________

Question 45 (JAM 2015 • Q47 • ID: JAM_2015_Q47) NAT +1 M Neg: Nil (0)

Q.7

If $5^{2015} \equiv n \pmod{11}$ and $n \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then n is equal to _________

MA 14/16

Question 46 (JAM 2015 • Q43 • ID: JAM_2015_Q43) NAT +1 M Neg: Nil (0)

Q.3

The number of distinct normal subgroups of S_3 is ______________

Question 47 (JAM 2015 • Q32 • ID: JAM_2015_Q32) **MSQ** **+2 M** **Neg: Nil (0)**

Q.2 Which of the following statements is (are) true?

(A) $\mathbb{Z}_2 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_6$

(B) $\mathbb{Z}_3 \oplus \mathbb{Z}_3$ is isomorphic to $\mathbb{Z}_9$

(C) $\mathbb{Z}_4 \oplus \mathbb{Z}_6$ is isomorphic to $\mathbb{Z}_{24}$

(D) $\mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5$ is isomorphic to $\mathbb{Z}_{30}$

MA 11/16

Question 48 (JAM 2015 • Q13 • ID: JAM_2015_Q13) **MCQ** **+2 M** **Neg: −0.67**

Q.13 Let G be a nonabelian group. Let $\alpha \in G$ have order 4 and let $\beta \in G$ have order 3. Then the order of the element $\alpha\beta$ in G

(A) is 6
(B) is 12
(C) is of the form $12k$ for $k \geq 2$
(D) need not be finite

MA 7/16

Question 49 (JAM 2015 • Q11 • ID: JAM_2015_Q11) **MCQ** **+2 M** **Neg: −0.67**

Q.11 Let S_3 be the group of permutations of three distinct symbols. The direct sum $S_3 \oplus S_3$ has an element of order

(A) 4
(B) 6
(C) 9
(D) 18

Question 50 (JAM 2015 • Q1 • ID: JAM_2015_Q1) **MCQ** **+1 M** **Neg: −0.33**

Q.1 Suppose N is a normal subgroup of a group G. Which one of the following is true?

(A) If G is an infinite group then G/N is an infinite group
(B) If G is a nonabelian group then G/N is a nonabelian group
(C) If G is a cyclic group then G/N is an abelian group
(D) If G is an abelian group then G/N is a cyclic group

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Groups (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	**9**
2	NAT	+1	**4**
3	NAT	+1	**90**
4	MSQ	+2	**A, B**
5	MCQ	+2	**A**
6	MCQ	+2	**C**
7	MCQ	+2	**C**
8	MCQ	+2	**C**
9	MCQ	+2	**A**
10	MCQ	+1	**B**
11	NAT	+2	**3 to 3**
12	NAT	+1	**3 to 3**
13	MSQ	+2	**C;D**
14	MCQ	+2	**C**
15	MCQ	+2	**D**
16	MCQ	+1	**B**
17	NAT	+2	**4 to 4**

Q. No.	Type	Marks	Official Key
18	NAT	+2	**31 to 31 or 46 to 46**
19	NAT	+1	**99 to 99**
20	MSQ	+2	**B, C**
21	MSQ	+2	**A, C**
22	MCQ	+2	**D**
23	NAT	+1	**0 to 0**
24	NAT	+1	**4 to 4**
25	MSQ	+2	**B;C;D**
26	MSQ	+2	**A;D**
27	MCQ	+2	**D**
28	MCQ	+2	**C**
29	MCQ	+2	**A**
30	MCQ	+1	**B**
31	NAT	+2	**29.9 to 30.1**
32	NAT	+1	**7.9 to 8.1**
33	NAT	+1	**−0.01 to +0.01**
34	NAT	+1	**5.9 to 6.1**

Q. No.	Type	Marks	Official Key
35	MSQ	+2	**A, C**
36	MSQ	+2	**A, C**
37	MCQ	+1	**B**
38	NAT	+2	**3.0 to 3.0**
39	NAT	+1	**6.0 to 6.0**
40	MSQ	+2	**B;C**
41	MCQ	+2	**B**
42	MCQ	+2	**D**
43	MCQ	+1	**B**
44	NAT	+2	**84**
45	NAT	+1	**1**
46	NAT	+1	**3**
47	MSQ	+2	**A;D**
48	MCQ	+2	**D**
49	MCQ	+2	**B**
50	MCQ	+1	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.